

ISBA 4775-01: Networking & Cloud Computing

Loyola Marymount University | Fall 2026

Course	ISBA 4775-01, Networking & Cloud Computing
Credit hours	4
Meets	Tuesday and Thursday, 1:45 PM to 3:25 PM
Location	Hilton 115

Instructor

Instructor	Greg Lontok, Clinical Associate Professor
Office	Hilton 114
Preferred contact	Microsoft Teams message, for the fastest response
Course Teams channel	https://r.isba.co/networking-cloud-teams-fa26
Email	gregory.lontok@lmu.edu
Office hours	By appointment. Book at https://calendly.com/greg-lontok

Course description

This course will introduce students to the fundamentals of networking and cloud computing. Students will configure networks and cloud computing services to address common information systems and business analytics needs through hands-on exercises. The course will start with networking fundamentals covering topics such as networking hardware, the OSI Model, TCP/IP, and various network protocols, while addressing each topic's security considerations. The second half of the course will leverage the student's networking foundation to explore and deploy the most commonly used cloud services, including compute, storage, databases, and serverless. Students will be able to host cloud-based applications and know the difference between the various cloud services available. This course may not be repeated for degree credit.

Prerequisites

ISBA 3710 with a minimum grade of C-.

What this course is for

You already know how to code, and this course is about what happens after that: getting something you built to run somewhere other than your laptop, keeping it running, and being able to explain how it works to someone who asks hard questions.

By the end you should be able to walk into an unfamiliar system and make sense of it.

What you own

You will use AI throughout this course, heavily, and that is expected.

What stays yours is the system. You know how it works, how it talks to other systems, what it is for, and how it delivers value to the stakeholder. It has to be fit for purpose: not only that you can build one, but that it suits the environment and constraints it has to live in.

Two whiteboard interviews, together worth 42.5% of your grade, are where that gets observed. You diagram your own system from memory, with no notes and no repository open, and then answer for it. "Working with AI" below is the full policy.

Learning outcomes

By the end of this course you can:

1. Read an unfamiliar system. Identify its components, dependencies, and boundaries, and ask the questions that reveal purpose, alternatives, cost, and where the pieces are isolated or coupled.
2. Design a system fit for purpose. Select and justify infrastructure against a stakeholder's real environment, constraints, and budget, and defend why that choice differs for a hobby project, a startup, and an enterprise.
3. Build and deploy an application on cloud infrastructure, moving it deliberately through local development, staging, and production.
4. Troubleshoot a system under failure. Form and test hypotheses, isolate the fault along the dependency chain, and describe what went wrong in the language practitioners use.
5. Operate a deployed system responsibly for access, security, performance, and cost, and produce evidence it is working as intended.
6. Explain and defend the system to a stakeholder. What it does, why it was built this way, which alternatives were rejected, where its limits are, and what should happen next.

Grading

Component	Points
Exercises (10 × 15)	150
Project 1, Own Your Corner of the Internet	125
Midterm whiteboard interview	125
Project 2, the job-description-driven build	300
Final whiteboard interview	300
Total	1000

Projects are 42.5% and interviews are 42.5%. This course grades whether you can explain what you built about as heavily as the artifact itself.

Letter grades

Grade	Range		Grade	Range
A	93-100		C+	77-79
A-	90-92		C	73-76
B+	87-89		C-	70-72
B	83-86		D	60-69
B-	80-82		F	below 60

Exercises, 150 points

There are ten exercises, worth 15 points each. Exercise 01 and Exercise 02 are submitted as files through Brightspace. Later build exercises use a public portfolio repository as specified in their briefs. Each brief defines its required evidence and grading.

Exercise 02 extends the in-class server lab: independently investigate a stopped Nginx service and a stopped MySQL service, explain your evidence, repair each service, and repeat the failed test. Submit a hand-drawn system diagram as a JPG and a short investigation report as Markdown. The rubric is 8 points for the two investigations, 4 for the diagram and explanation, and 3 for the healthy baseline and verified Codespace shutdown.

Exercise schedule

#	Exercise	Type	Due
01	Accounts, your home network, and the self-discovery interview	Config	Tue 9/8
02	Troubleshoot a cloud service: Nginx and MySQL	Troubleshoot	To be announced
03	The AI-assisted workflow: clarify, plan, build, verify	Build	To be announced
04	Domain delegated to Route 53, live with TLS	Config	Thu 9/24
05	Broken DNS/TLS: diagnose and fix	Troubleshoot	Thu 10/1
06	Mailbox email with SPF, DKIM, DMARC; Resend verified	Config	Thu 10/8
07	First orchestration: trigger, API call, notification	Build	Thu 10/15
08	Broken integration: expired credential, changed schema	Troubleshoot	Tue 10/20
09	Raw agent loop in Python, three tools, evals, trace read	Build	Thu 11/5
10	Broken agent: bad tool schema, runaway loop, cost blowup	Troubleshoot	Thu 11/12

Ex02's proposed date is September 15; confirm the final deadline in Brightspace. Ex03's final scope and deadline are being revised. The personal site, Railway deployment, and GA4 remain Project 1 requirements; their place in the exercise sequence will be announced with the revised Ex03 brief.

Project 1: Own Your Corner of the Internet, 125 points

Project 1 runs from week 1 through week 8, and it is **due Thu 10/22**.

The exercises build the pieces. Project 1 is the integrated, documented, portfolio-grade assembly of them.

Deliverable	Requirement
Domain	Purchased, delegated to Route 53, hosted zone under your control
Site	Personal site on Railway, live at your domain with valid TLS
CI/CD	GitHub Actions, push to main deploys
Mailbox email	Zoho at your domain; SPF, DKIM, DMARC published and passing
Transactional email	Resend verified on your domain
Analytics	GA4 installed and reporting
Job Scout	An orchestration that finds local companies, reads their careers pages, scores openings against your resume, and emails you a digest from your own domain
Architecture diagram	One page, labeled, every service and the data flow between them
README	Configuration guide, an FAQ of at least five questions drawn from real failures, and a retrospective

The question is not whether you completed the steps but whether you would show this repository to an employer.

Your Job Scout must surface at least five real postings you would actually want. You pick one, and it becomes Project 2's brief. If your scout underperforms, there is a vetted feed of postings so you are not blocked.

Midterm whiteboard interview, 125 points

The midterm is twenty minutes, in person and one-on-one, scheduled through Calendly. Slots run **Tue 10/27 through Thu 10/29**, and a study guide is published in advance.

You spend the first fifteen minutes on your own system. You diagram it from memory, with no notes and no repository open, and then you answer for it: why this service, what you rejected, what it costs, how you know it works, what breaks first under load, and what you would do differently.

The last five minutes are a curveball. I change one constraint, drawn from a deck whose categories you have seen but whose specifics you have not. Traffic might increase tenfold, the budget might be cut by eighty percent, or a vendor might sunset a service you depend on. You redesign on the spot.

Most students have never done a whiteboard interview. The October one is where you find out what that is like while it still costs 125 points instead of 300, and that is why it sits before the November 13 withdrawal deadline.

Project 2: the job-description-driven build, 300 points

Project 2 runs from week 10 through week 15. You pick one posting your Job Scout surfaced, read the system that job implies, and build it. The architecture is fixed, and the job description chooses the problem domain.

Deliverable	Requirement
Web app	On a subdomain of the domain you already own
Agent	Custom-built: tools, memory, structured outputs, guardrails, embedded as an endpoint
Grounding	Domain data in S3, reached with least-privilege IAM credentials
Orchestration	A layer connecting two systems on a trigger, doing real work in the job's domain
Environments	Local, staging, production, with promotion through CI/CD
CI/CD	GitHub Actions, secrets managed properly, deploy on merge
Operations	Auth, rate limiting, analytics, cost controls, a monitoring story
Evidence	An eval set for the agent and an inspectable trace
Decision record	Your most important decision, the evidence, how AI contributed, what you accepted or rejected or changed, how you validated it, your biggest remaining uncertainty, and what is next
Blog post	The writeup

Everyone builds the same skeleton onto a different problem. A logistics posting might yield a shipment-status copilot, a marketing posting a campaign analyzer, and a healthcare-operations posting an intake triage assistant.

The orchestration requirement is a capability, not a tool. n8n, Make, Zapier, Power Automate, and code all count.

Your system has to be both deterministic and agentic, and you have to defend where the line between them falls. Knowing which parts should be a fixed workflow and which should be agentic is a design judgment, and it is the obvious curveball: your API costs tripled, so what moves out of the agent?

Milestones

These are gates inside the Project 2 grade rather than separate line items.

Milestone	Bar	Due
M1	Posting selected, PRD written, architecture diagram approved	Tue 11/10
M2	App scaffolded and deployed to production on your subdomain	Tue 11/17
M3	Agent responding with tools, grounded in domain data via S3 and IAM	Tue 11/24
M4	Orchestration integrated, staging environment live and promoting to production	Thu 12/3
Final	Full system, operations hardened, evals, decision record, blog post	Tue 12/8

M4 fails without a working staging environment. The application runs fine without one, which makes it the first thing skipped under deadline pressure.

Final whiteboard interview, 300 points

Thursday, December 17, 11:00 AM, in the scheduled exam slot, for thirty minutes.

Same format as the midterm: twenty-five minutes on your own system from memory, five minutes of curveball. The system you defend is the one you built for a job you chose, so this is a simulated interview for that job.

Semester at a glance

Weeks	What happens
1-8	Networking foundations, then the cloud arc. Exercises 01-08, Project 1
9	Midterm whiteboard interviews
10-15	Project 2, the job-description-driven build. Exercises 09-10, milestones M1-M4
Finals	Final whiteboard interview

Thanksgiving week meets once, on Zoom. Tuesday, November 24 is remote, and you join from the Zoom tool in Brightspace rather than coming to Hilton 115. Thursday, November 26 is Thanksgiving Day, and it is the only cancelled meeting all semester.

Labor Day and Autumn Day fall on days this course doesn't meet.

Key dates

Event	Date
Add/drop without a W	Fri Sep 4
Project 1 due	Thu 10/22
Midterm interviews	Tue 10/27 to Thu 10/29
Last day to withdraw	Fri Nov 13
Instruction ends	Fri Dec 11
Final interview	Thu Dec 17, 11:00 AM

Class schedule

Date	Session	Due
Tue 9/1	The map: where a request goes, and the devices it passes through	
Thu 9/3	Finding things: host names, IP and MAC addresses, the default gateway, DNS, and what your own machine will tell you	
Tue 9/8	Ports and web services: requests, responses, HTTP status codes, and investigating a service failure	Ex01
Thu 9/10	Running and Troubleshooting Services: physical server, VM, container; Nginx and MySQL; processes, ports, logs; database creation, tables, and sample data	
Tue 9/15	Database to application: portfolio from a template, repeatable setup, Python build-along, environment variables and Codespaces secrets, failure/recovery, commit/push and a small change; optional instructor PaaS demo	Ex02 proposed; confirm in Brightspace
Thu 9/17	DNS from the root down: zones, records, TTL, delegation, and nameservers	

Date	Session	Due
Tue 9/22	TLS: what a certificate claims, how a browser decides it is valid, the four ways it goes invalid, and how ACME gets you one	
Thu 9/24	Reading a system when it breaks: dig, curl, openssl, and the hypothesis log	Ex04
Tue 9/29	Failures that depend on who is asking: DNS propagation and TTL, cached answers, and why an incomplete chain passes in a browser but fails in curl	
Thu 10/1	Diagnosis walkthroughs, then how mail routes: MX records and the envelope	Ex05
Tue 10/6	Email authentication: SPF, DKIM, DMARC, and deliverability as a DNS problem	
Thu 10/8	APIs: REST, JSON, status codes, authentication, rate limits, and idempotency	Ex06
Tue 10/13	Orchestration: triggers, webhooks, polling, and joining two systems	
Thu 10/15	How integrations break: expired credentials, changed schemas, and reading 401, 403, 429, and 5xx	Ex07
Tue 10/20	CI/CD with GitHub Actions, secrets, deploy on merge, and architecture diagrams	Ex08
Thu 10/22	Project 1 walkthroughs, the midterm study guide, and what a whiteboard interview is	Project 1
Tue 10/27	Midterm whiteboard interviews, no class session	
Thu 10/29	LLM calls: tokens, context windows, cost, and structured outputs	
Tue 11/3	The agent loop: tools, schemas, memory, and reading a trace	
Thu 11/5	Evals: what to measure, building a small eval set, and finding why a run failed	Ex09
Tue 11/10	Reading a job description as a system spec, and architecture review one by one	M1
Thu 11/12	Guardrails: loop limits, schema validation, token budgets, and cost controls	Ex10
Tue 11/17	Environments: local, staging, production, and promotion through CI/CD	M2
Thu 11/19	Cloud storage and databases: object versus relational, S3, and least-privilege IAM	
Tue 11/24	Grounding review, retrieval, chunking, and what the agent must not see. On Zoom	M3
Thu 11/26	No class, Thanksgiving	

Date	Session	Due
Tue 12/1	Operating a deployed system: auth, rate limiting, monitoring, and cost controls	
Thu 12/3	Staging in practice, then the decision record	M4
Tue 12/8	Blog post workshop and final interview prep	Project 2
Thu 12/10	Final interview practice in pairs, and a course retrospective	

September 15 continues the unfinished application build from September 10. Topics from September 17 onward remain the planned sequence. Ex03 and the personal-site build are being rescheduled; any affected lesson or deadline changes will be announced in class and Brightspace.

Work load expectations

LMU follows the Carnegie Unit standard: one semester credit hour represents at least three hours of student engagement per week. See <https://academics.lmu.edu/aprc/lmucredithourpolicy/>

For this four-credit course that is an average of **12 hours per week** across the semester, including class meetings, exercises, project work, and preparation.

Expect this to be uneven. Weeks with a troubleshooting exercise and weeks around a Project 2 milestone run heavier than the middle of a build.

Required materials

There is no textbook. This course runs on real infrastructure, which costs real money, and most of the spending starts in Exercise 01.

What	Cost	When
Claude Pro or an accepted ChatGPT subscription	Claude Pro is \$20/month on monthly billing; alternative plan pricing varies	Choose one in Ex01; follow the relevant assignment for coding-agent setup
Railway	Trial credit is limited; Hobby has a \$5/month minimum with usage-based billing	Personal-site deployment for Project 1; setup timing to be announced
Domain name	Free for a year through the GitHub Student Developer Pack, otherwise \$10 to \$15	Exercise 04, and it stays yours after the course
Route 53 hosted zone	\$0.50 a month	Exercise 04
GitHub	Free tier	Exercise 01
GitHub Codespaces	Personal-account compute/storage allowance; additional usage can cost money	Course Codespace in Session 04 for Ex02; new portfolio Codespace in Session 05
AWS	Free tier	Exercise 04

What	Cost	When
Zoho and Resend	Free tier	Exercise 06
Google Cloud and Firecrawl	Free tier	Exercise 07

Plan your budget around the AI subscription you choose, domain registration, and the services you actually run. Prices, credits, and eligibility can change. Check [Claude pricing](#), [Railway plans](#), and [Codespaces billing](#) before enabling paid usage. A free tier does not mean all usage is free.

Apply for the [GitHub Student Developer Pack](#) in week 1 and check the current offers and verification status in your account. Do not plan on a particular benefit until you have confirmed eligibility.

For the Codespaces lab, check your allowance with the instructor, keep forwarded ports private, and stop the environment when finished. Contact me if access is blocked or a billing change is requested. Bring cost questions to me early.

Required lab fees

None.

Instructional methods

We meet in person twice a week, and each session combines instruction, live work, and student walkthroughs of their own exercises. Guided setup and build-alongs include predictions, live troubleshooting, and brief discussion. Detailed incident reports and diagrams are completed independently in the assigned homework. There are also two one-on-one whiteboard interviews.

Assignments and feedback

Follow each assignment's submission instructions. Ex01 and Ex02 go to Brightspace as separate files. Later builds use repositories and live systems as specified in their briefs; projects include their repositories and deployed systems. Due dates are announced in Brightspace and reflected in this syllabus. A date marked proposed or to be announced is not a confirmed deadline.

For Ex02, Google Docs saves your report automatically. Confirm it is saved to Drive, stop the Codespace, add your shutdown note, and download the report as Markdown for submission. Your diagram is a separate JPG. The Codespace should be stopped after testing; it does not need to remain running for grading.

Feedback is given in class, in Brightspace, or in the repository, depending on the assignment. Interview feedback is given verbally at the end of the interview.

Deadlines are hard. Late work loses 10% per day, down to a floor of 50%. A late assignment is worth less, but it's never worth nothing, so finish it and turn it in.

If you know you're going to miss a deadline, send me a Teams message before it passes. I may move it. That's case by case, and there's no bank of late days to spend. If I move a deadline, the new date is the deadline, and the 10% per day runs from there. No message before the deadline means no conversation after it, and the 10% per day runs from the original date.

Some assignments are delivered in person at a fixed time, and the meeting time is the deadline. The two whiteboard interviews can't be rescheduled, so missing one scores zero on it.

Emergencies sit outside all of this. Tell me when you're able and we'll make a plan.

Attendance

Your scheduled commitments are class sessions, your midterm interview slot, and the final interview.

If you feel ill, stay home. Rest, and keep everyone else healthy.

When you can't make one, send me a Teams message. Beforehand is best, and as soon as you can if something happens that morning. Late still beats silent.

Your first three absences cost you nothing if you told me. You don't owe me a reason and I won't ask for documentation.

An absence you never tell me about costs 2% of your total grade, and it still uses up one of the three. From the fourth on, each absence costs 2% whether you told me or not. That 2% comes off your final grade after everything else is totaled, so attendance is not a line in the grading table.

Emergencies sit outside all of this. Serious illness, hospitalization, a death in the family: the three-absence count doesn't apply. Tell me when you're able and we'll work out a plan for the rest of the term.

Use of technology

We introduce tools when the relevant lesson or assignment needs them. In Session 04, students create a Codespace directly from the course repository. They install services and create a MySQL database with sample data.

In Session 05, students create their own public portfolio from the [instructor template](#). They add a Codespaces secret and open a new Codespace from their portfolio. GitHub clones that repository automatically. Setup scripts and SQL recreate the services and data inside its Linux container.

The coding agent interviews students, then helps build a Python application after plan approval. Students verify database reads, failure, and recovery. They commit and push a working version, then make and verify a small change before committing and pushing again. An optional instructor Railway demo connects this workflow to deployment. Student deployment instructions follow with the personal-site work.

The early server lab does not require a local editor, database, or coding-agent installation. The Ex03 brief will specify any local workflow setup. AWS, Railway, and orchestration tools are introduced with their later assignments.

See "Working with AI" below for how AI use is expected and assessed here.

For technology assistance, contact the ITS Service Desk:

- Phone: 310-338-7777 or 213-736-1097
- Email: servicedesk@lmu.edu
- Self-service: <https://its.lmu.edu/servicedesk>

Working with AI

Use AI for everything it is good at: planning, writing code, debugging, reading unfamiliar documentation, explaining errors, and drafting your PRDs.

There is no restriction on how much you use it and no requirement to log prompts.

What matters is what remains yours:

- You defend the system. An AI can build a system you cannot explain. The whiteboard interviews find that out.
- You validate your own work. Live systems, evals, traces, and an honest account of what you have not verified.
- You own the decisions. Which service, which alternative you rejected, where the deterministic/agentive line falls, and why. If AI recommended something and you took it, you should be able to say why it was right.
- You own what you ship, including its limits and what you would do next.

Independent work is not the same as independent thinking. You can use AI heavily and demonstrate excellent ownership. You can also work entirely alone and demonstrate very little.

Student support

- Academic Resource Center: <https://academics.lmu.edu/arc/>
- Student Psychological Services: <https://studentaffairs.lmu.edu/wellness/studentpsychologicalservices/>
- Disability Support Services: <https://academics.lmu.edu/dss/>
- Community of Care: <https://studentaffairs.lmu.edu/lmucare/>

If something outside this course is affecting your ability to do the work, tell me.

University policies

Academic honesty

Loyola Marymount University is a community dedicated to academic excellence, student-centered education, and the Jesuit and Marymount traditions. As such, the University expects all members of its community to act with honesty and integrity at all times, especially in their academic work. Academic honesty requires that all members of the LMU community act with integrity, respect their own intellectual and creative work as well as that of others, acknowledge sources consistently and completely, act honestly during exams and on assignments, and report results accurately. As an LMU Lion, by the Lion's Code, you are pledged to join the discourse of the academy with honesty of voice and integrity of scholarship.

Academic dishonesty will be treated as an extremely serious matter, with serious consequences that can range from receiving no credit for assignments/tests to expulsion. It is never permissible to turn in any work that has been copied from another student or copied from a source (including Internet) without properly acknowledging/citing the source. It is never permissible to work on an assignment, exam, quiz or any project with another person unless your instructor has indicated so in the written instructions/guidelines. It is your responsibility to make sure that your work meets the standard of academic honesty set forth in the "Academic Honesty Policy" found at: <https://academics.lmu.edu/honesty/> For an additional resource, see and the "LMU Honor Code and Process" found at: https://bulletin.lmu.edu/content.php?catoid=1&navoid=18#LMU_Honor_Code_and_Process

What this means in this course

AI use is expected and encouraged here, on the terms in "Working with AI" above. Submitting a system you cannot account for is a violation regardless of what produced it.

Two more violations are specific to this course. The first is presenting a system as working without evidence that it works. Report actual observations, including failed checks and unresolved problems; do not fabricate output. Evidence may be a private report, a repository, or a live demonstration, as specified in the assignment. The second is presenting someone else's system, drawing, or evidence as your own. You must be able to explain what you submit.

Special accommodations

The DSS Office offers resources to enable students with physical, learning, ADD/ADHD, psychiatric disabilities and those on the autism spectrum to achieve maximum independence while pursuing their educational goals. Staff specialists interact with all areas of the University to eliminate physical and attitudinal barriers. Students must provide documentation for their disability from an appropriate licensed professional. Services are offered to students who have established disabilities under state and federal laws. We also advise students, faculty and staff regarding disability issues. Students who need reasonable modifications, special assistance, academic accommodations or housing accommodations should direct their request to the DSS Office as soon as possible. All discussions will remain confidential. The DSS Office is located on the 2nd floor of Daum Hall and may be reached by email at dsslmu@lmu.edu or phone at (310) 338-4216. Please visit <http://www.lmu.edu/dss> for additional information.

The whiteboard interviews can be accommodated. The earlier I know, the better I can plan them.

Tentative nature of the syllabus

This syllabus and its contents are subject to revision; students are responsible for any changes or modifications announced or distributed in class or posted on LMU's course management system.

Exercise scope and dates may shift as the semester develops. Any revision will be announced in class and through Brightspace, the same channels through which this syllabus was distributed. The version at <https://r.isba.co/networking-cloud-syllabus-fa26> is always current, and its revision history is visible in git.

Classroom conduct and communication

Treat your classmates with professional respect, particularly when walking through each other's work. Much of this course involves showing something broken in front of the room, which only works if that is a safe thing to do.

Check your lion.lmu.edu email and Microsoft Teams regularly. Those are the channels I use and you are responsible for what is sent through them.

LMU's expectations for classroom behavior are at <https://lmu.box.com/s/v2x89uspgbx3l23egcz7mjd6dbekcn60>

Reporting requirements for sexual or interpersonal misconduct: <https://studentaffairs.lmu.edu/lmucares/>

Emergency preparedness information: <https://www.lmu.edu/emergency>

College of Business Administration mission

We advance knowledge and develop business leaders with moral courage and creative confidence to be a force for good in the global community.

<https://cba.lmu.edu/about/mission/>